Magnetic Fields and Electromagnetism Worksheet

1. A small bar magnet is hidden in a fixed position inside a tennis ball. Describe an experiment you could do to find the location of the North pole and the South pole of the magnet.

2. A piece of metal is attracted to one pole of a large magnet. Describe how you could tell whether the metal is a temporary magnet of a permanent magnet.

3. A wire 0.50 m long carrying a current of 8.0 A is at right angles to a 0.40 T magnetic field. How strong a force acts on the wire?

4. A wire 75 cm long carrying a current of 6.0 A is at right angles to a uniform magnetic field. The magnitude of the force acting on the wire is 0.60 N. What is the strength of the magnetic field?

5. A copper wire 40 cm long carries a current of 6.0 A and weighs 0.35 N. A certain magnetic field is strong enough to balance the force of gravity on the wire. What is the strength of the magnetic field?

6. A wire 1.5 m long carrying a current of 10.0 A is at right angles to a uniform magnetic field. The force acting on the wire is 0.60 N. What is the strength of the magnetic field?

7. A wire 0.50 m long carrying a current of 8.0 A is at right angles to a uniform magnetic field. The force on the wire is 0.40 N. What is the strength of the magnetic field?

8. The current though a wire 0.80 m long is 5.0 A and points towards the west. The wire is perpendicular to a 0.60 T magnetic field that is pointing to the North. What is the magnitude and direction of the force on the wire?

9. A wire 35 cm long is parallel to a 0.53 T uniform magnetic field. The current though the wire is 4.5 A. What force acts on the wire?

10. A wire 625 m long is in a 0.40 T magnetic field. A 1.8 N force acts on the wire. What is the magnitude of the current flowing through the wire?

11. The force acting on a wire at right angles to a 0.80 T magnetic field is 3.6 N. The current the wire is 7.5 A. How long is the wire?

12. A power line carries a 225 A current from east to west parallel to the surface of the Earth. The Earth's magnetic field is $5.0\text{x}10^{-5}$ T (and it points towards the north). Calculate the magnitude and direction of force acting on the wire.